Dikshant Lamsal

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Education

BSc in Mechanical Engineering | Fall 2024 - Present | Louisiana Tech University, USA

GPA: 3.94/4.00

Technical Skills:

CAD and Design: SolidWorks (part modeling, assemblies, DXF export, basic drawings), tolerance awareness.
Simulation and Analysis: MathCAD (Kinematics, structural calculations), basic FEA exposure, stiffness and deflection reasoning.
Prototyping and Fabrication: FDM 3D printing, mechanical assembly, fasteners, component integration
Mechanical Systems: Structural truss optimization, small vehicle systems, thermal control systems
Programming: Arduino (C-based), Python

Projects

Senses Lamp:

Co-designed and built the "Senses Lamp", a smart home device with sunrise simulation, adjustable lighting, alarm, Bluetooth speaker, phone charging, and environmental displays.

Modeled base and neck assembly in SolidWorks to house electronics with attention to structural stability and internal component layout.

Neck Posture Corrector:

Designed and developed a wearable posture correction system using Arduino and a flex sensor to detect neck bending and poor posture.

Implemented a vibration-based feedback mechanism to provide real time, non-intrusive alerts, improving user awareness and posture habits. Prototyped compact housing in Solidworks and integrated the system with a glasses-based setup for practical, everyday use.

Nuclear Power Plant Mapping:

Collaborated with Dr. Chris Jacobsen (Research Professor, Northwestern University) on data collection for a book on energy, climate, and nuclear power.

Used Python (Spyder) to process data and support creation of a copyright-free global map with reactor sites sized by capacity and color coded by decade.

Truss Design and Optimization:

Designed a structural truss in SolidWorks, created DXF parts. 3D printed individual components (FDM), considering layer orientation and fit tolerances.

Assembled all parts by hand and optimized members using I and T shaped cross sections to increase strength while minimizing weight.

Arduino Based Smart Car:

Built a small Arduino-powered car that responds to directional inputs using photoresistors and gained hands on experience in Arduino coding, sensor integration, and basic robotics. Diagnosed and corrected drivetrain alignment and sensor response issues during testing.

Additional Experience

IT Support Specialist | Orbit English School, Nepal | (Fall 2021-Spring 2022)

Maintained and troubleshot computer lab hardware, including desktops, monitors, and peripherals, to ensure smooth operations.

Diagnosed and resolved technical issues, providing timely support to students and staff for both hardware and software problems.

Activities:

3rd Runner up in Integration Bee 2025 at Louisiana Tech University

Public Relation Officer, International Student Association (2025/26)